planet. Those on highly elliptical trajectories, those that cross the orbits of other planets, will sooner or later collide or, by a near miss, be gravitationally ejected from the Solar System.

The planets almost certainly accumulated from worldlets which in turn had condensed out of a great flat cloud of gas and dust surrounding the Sun—the sort of cloud that can now be seen around young nearby stars. So, in the early history of the Solar System before collisions cleaned things up, there should have been many more worldlets than we see today.

Indeed, there is clear evidence for this in our own backyard: If we count up the interloper worldlets in our neighborhood in space, we can estimate how often they'll hit the Moon. Let us make the very modest assumption that the population of interlopers has never been smaller than it is today. We can then calculate how many craters there should be on the Moon, The number we figure turns out to be much less than the number we see on the Moon's ravaged highlands. The unexpected profusion of craters on the Moon speaks to us of an earlier epoch when the Solar System was in wild turmoil, churning with worlds on collision trajectories. This makes good sense, because they formed from the aggregation of much smaller worldlets—which themselves had grown out of interstellar dust. Four billion years ago, the lunar impacts were hundreds of times more frequent than they are today; and 4.5 billion years ago, when the planets were still incomplete, collisions happened perhaps a billion times more often than in our becalmed epoch.

The chaos may have been relieved by much more flamboyant ring systems than grace the planets today. If they had small moons in that time, the Earth, Mars, and the other small planets may also have been adorned with rings.

The most satisfactory explanation of the origin of our own Moon, based on its chemistry (as revealed by samples returned from the *Apollo* missions), is that it was formed almost 4.5 billion years ago, when a world the size of Mars struck the Earth. Much of our planet's rocky mantle was reduced to dust and hot gas and blasted into space. Some of the debris, in orbit around the Earth, then gradually reaccumulated—atom by atom, boulder by boulder. If that unknown